DIRECTIONS *for questions 1 to 4*: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

The Bronze Age in China, which began around 2000 B.C., saw the growth of a civilization sustained in its essential aspects for another 2,000 years. The area along the Yellow River became the seat of the political and military power of the Shang dynasty (ca. 1600–1046 B.C.).

The Shang dynasty was conquered by the people of Zhou, who came from farther up the Yellow River in the area of Xi'an in Shaanxi Province. In the first years of the Zhou dynasty (1046–256 B.C.), known as the Western Zhou (1046–771 B.C.), the ruling house of Zhou exercised a certain degree of "imperial" power over most of central China. The second phase of the Zhou dynasty, known as the Eastern Zhou (770–256 B.C.), is subdivided into two periods, the Spring and Autumn period (770–ca. 476 B.C.) and the Warring States period (475–221 B.C.). During the Warring States period, seven major states contended for supreme control of the country, ending with the unification of China under the Qin in 221 B.C.

The era of the Shang and the Zhou dynasties is generally known as the Bronze Age of China, because bronze, an alloy of copper and tin, used to fashion weapons, parts of chariots, and ritual vessels, played an important role in the material culture of the time. Iron appeared in China toward the end of the period, during the Eastern Zhou dynasty.

The earliest Chinese bronzes were made by the method known as piece-mold casting—as opposed to the lost-wax method, which was used in all other Bronze Age cultures. In piece-mold casting, a model is made of the object to be cast, and a clay mold taken of the model. The mold is then cut in sections to release the model, and the sections are reassembled after firing to form the mold for casting. If the object to be cast is a vessel, a core must be placed inside the mold to provide the vessel's cavity. An advantage of this rather cumbersome way of casting bronze was that the decorative patterns could be carved or stamped directly on the inner surface of the mold before it was fired. This technique enabled the bronzeworker to achieve a high degree of sharpness and definition in even the most intricate designs.

One of the most distinctive and characteristic images decorating Shang-dynasty bronze vessels is the so-called taotie. This frontal animal-like mask has a prominent pair of eyes, often protruding in high relief. Other common motifs for Shang ritual bronze vessels were dragons, birds, bovine creatures, and a variety of geometric patterns. Currently, the significance of the taotie, as well as the other decorative motifs, in Shang society is unknown.

Jade, along with bronze, represents the highest achievement of Bronze Age material culture. Shang craftsmen had full command of the artistic and technical language developed in the diverse late Neolithic cultures that had a jade-working tradition. If the precise function of ritual jades in the late Neolithic is indeterminate, such is not the case in the Bronze Age. Written records and archaeological evidence inform us that jades were used in sacrificial offerings to gods and ancestors, in burial rites, for recording treaties between states, and in formal ceremonies at the courts of kings.

Q1. Which of the following conclusions CANNOT be drawn from the data presented in the last para of the passage?

a) There is no evidence of Neolithic era use of ritual jades.

b) There are written records for the ritualistic use of jade in the Shang society.

c) Jade had many official uses even during the Bronze Age.

d) **The Neolithic era preceded the Bronze Age.** ✘ Your answer is incorrect

Q2. Which of the following, if answered, will most likely add further depth to the discussion about the Bronze Age in China?

- a) Whether the purpose of the taotie and other decorative motifs used during Bronze Age went beyond ornamentation
- b) **Whether the Qin dynasty encouraged bronze artifacts as much as the Shang and Zhou dynasties did** ✗ **Your answer is incorrect**
- c) Whether jade was a more expensive material compared to bronze
- d) **Whether the Neolithic era craftsmen preferred working with jade over bronze**

Q3. All of the following are true according to the passage EXCEPT:

- a) **Iron appeared in China only after 800 BC.** ✗ Your answer is incorrect
- b) **The Bronze Age was named so because of the predominance of the usage of bronze in war and rituals alike.**
- c) A less popular method of bronze-making was used by Chinese bronze-workers to achieve greater sharpness and definition.
- d) **Bronze craftsmen honed the technique and skills associated with jade-working traditions.**

Q4. Which of the following sequences correctly represents the dynastic periods of rule mentioned in the passage?

- a) **Shang—Zhou—Spring—Autumn—Warring States and Qin**
- b) **Shang—Western Zhou—Eastern Zhou—Warring States—Qin** ✘ **Your answer is incorrect**
- c) Shang—Western Zhou—Spring and Autumn—Warring States—Qin
- d) **Shang—Zhou—Spring and Autumn—Warring States—Qin**

DIRECTIONS *for questions 5 to 8*: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Recent data points that surfaced in China, then Europe, then all over, point to a picture of a brief, cataclysmic hot spell 56 million years ago, now known as the Paleocene-Eocene Thermal Maximum (PETM). After heat-trapping carbon leaked into the sky from an unknown source, the planet, which was already several degrees Celsius hotter than it is today, gained an additional 6 degrees. The ocean turned jacuzzi-hot near the equator and experienced mass extinctions worldwide. On land, primitive monkeys, horses and other early mammals marched northward, following vegetation to higher latitudes. The mammals also miniaturized over generations, as leaves became less nutritious in the carbonaceous air. Violent storms ravaged the planet; the geologic record indicates flash floods and protracted droughts.

The PETM doesn't only provide a past example of CO2-driven climate change; scientists say it also points to an unknown factor that has an outsize influence on Earth's climate. When the planet got hot, it got really hot. Ancient warming episodes like the PETM were always far more extreme than theoretical models of the climate suggest they should have been. Even after accounting for differences in geography, ocean currents and vegetation during these past episodes, paleoclimatologists find that something big appears to be missing from their models – an X-factor whose wild swings leave no trace in the fossil record.

Evidence is mounting in favour of the answer that experts have long suspected but have only recently been capable of exploring in detail. "It's quite clear at this point that the answer is clouds," said Matt Huber, a paleoclimate modeler at Purdue University. Clouds currently cover about two-thirds of the planet at any moment. But computer simulations of clouds have begun to suggest that as the Earth warms, clouds become scarcer. With fewer white surfaces reflecting sunlight back to space, the Earth gets even warmer, leading to more cloud loss. This feedback loop causes warming to spiral out of control.

New findings reported today in the journal Nature Geoscience make the case that the effects of cloud loss are dramatic enough to explain ancient warming episodes like the PETM — and to precipitate future disaster. Climate physicists at the California Institute of Technology performed a state-of-the-art simulation of stratocumulus clouds, the low-lying, blankety kind that have by far the largest cooling effect on the planet... In the simulation, when the tipping point is breached, Earth's temperature soars 8 degrees Celsius, in addition to the 4 degrees of warming or more caused by the CO2 directly.

Once clouds go away, the simulated climate "goes over a cliff," said Kerry Emanuel, a climate scientist at the Massachusetts Institute of Technology. He noted, scientists must now make an effort to independently replicate the work. Huber said the stratocumulus tipping point helps explain the volatility that's evident in the paleoclimate record. "Schneider and co-authors have cracked open a Pandora's box of potential climate surprises," he said, adding that, as the mechanisms behind vanishing clouds become clear, "all of a sudden this enormous sensitivity that is apparent from past climates isn't something that's just in the past. It becomes a vision of the future."

Q5. The 'Pandora's box of potential climate surprises' mentioned in the last para of the passage points to all of the following EXCEPT:

a) **Disappearance of clouds because of environmental warming**

b) **Temperature volatility because of cloud loss**

c) Additional spike in temperature following an overall increase in average temperature of the environment

d) **Environmental warming because of extremely high global CO2 levels**

Q6. Which of the following best states the underlying assumption in Kerry Emanuel's suggestion in the last para?

- a) **New findings reported in Nature Geoscience have not yet been reviewed by other scientists.**
- b) **Multiple simulations need to be run by other scientists to confirm the drastic effects of cloud loss due to warming.**
- c) It needs to be verified whether there is a corelation between warming of the planet and cloud loss.
- d) **It needs to be confirmed whether the stratocumulus tipping point explains temperature volatility in the past.**

Q7. The PETM event was mentioned by the author to explain

- a) **why there was a cataclysmic hot spell 56 million years ago.**
- b) **that CO_2-driven climate change could cause cascading issues with disproportionate influence on earth's climate**
- c) that high CO_2 levels beyond the tipping point can directly cause an outsized influence on earth's climate.
- d) **that ancient warming episodes like the PETM were always far more extreme than theoretical models of the climate suggest**

Q8. The statement that "paleoclimatologists find that something big appears to be missing from their models" (para 2) leads to the conclusion that:

- a) **Theoretical models didn't account for all the factors affecting earth's climate.**
- b) **The amount of cloud cover during the PETM is hard to estimate.**
- c) Carbon acts as a heat trap that warms the environment disproportionately.
- d) **Clouds can be adversely affected by the increase in environmental temperatures.**

DIRECTIONS *for questions 9 to 12*: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Few fields of inquiry draw from as diverse an array of disciplines as the study of human fertility. Contributions come from sociology, public health, medicine, demography, political science, economics, and anthropology, each discipline bringing to bear its own particular perspectives and theoretical agenda. The perspective of the biological anthropologist among these others is unique in two respects: the central position of evolutionary theory and evolutionary history in anthropological thinking, and the commitment to understanding human fertility, its determinants and consequences, as part of an integrated species biology.

These two elements give rise to the two primary motivations of biological anthropologists for studying human fertility. The first motivation is to understand as fully as possible our evolutionary past, both the history of change that we and our phyletic relatives have undergone and the forces and constraints that have shaped its course. Understanding the reproductive biology of our species is fundamental to that effort, since in its essentials, evolution or natural selection can be broken down into variability in the processes of birth and death. If we can fully comprehend the way in which our fertility is regulated by physiological, ecological and social mechanisms, we will be in a better position to elucidate critical junctures of human evolution, such as the transition to subsistence horticulture, the transition to cooperative hunting and gathering societies, or even the original diversion from other hominoid lines.

The second motivation is in some ways the converse of the first: to understand our species in the present as the product of our formative evolutionary past. That is, as distinct from the often eidetic perspective of medicine or the mechanical perspective of physiology, the biological anthropologist views the apparatus and process of human fertility as shaped by the action of natural selection and searches for the logic of ultimate functionality that can provide a theoretical framework unifying this and other aspects of human biology.

As a result of these two motivations and the basic tenets of the discipline underlying them, the most important contribution of biological anthropology to the study of human fertility is theoretical cohesion, providing a connection to the overarching framework of evolutionary biology.

A quiet revolution has occurred in the last 30 years in our understanding of human fertility which has brought the perspective of biological anthropology to the forefront of the field. The previous view, directly descended from Thomas Malthus, postulated a constant and prolific biology manipulated and constrained by culture, with all the important determinants of variance in human fertility deriving from cultural conventions, societal pressures or individual choice. We now appreciate more fully the degree to which human reproductive physiology is naturally variable and human fertility biologically determined, and the extent to which that determination can be accounted for by evolutionary theory. Rather than the empirical quagmire that characterized the state of the field in the 1950s, we are now well on the way to a theory of natural human fertility. Such an organized theoretical perspective is necessary to support practical applications of knowledge in this area, as, for instance, in anticipating important policy issues in developing societies, or informing clinical perspectives on fertility and infertility. However, scientific knowledge can only support and inform practical decisions concerning policy and intervention. It can never dictate a particular course of action or validate a given policy…

Q9. It cannot be inferred from the first para of the passage that

a) **human fertility and human evolution are interrelated.**

b) **the perspective of biological anthropologists on human fertility is more important compared to those practising other disciplines.**

c) species biology is interconnected with anthropological thinking.

d) **other disciplines such as sociology, medicine, political science and economics aren't centred around evolutionary theory.** ✗ Your answer is incorrect

Q10. The author believes that comprehending the reproductive biology of our species is important to understand our evolutionary past because

- a) **our evolutionary past is nothing but the history of our change.**
- b) **understanding the forces and constraints that were imposed on evolution is imperative for understanding evolution.**
- c) reproductive biology influences the process of natural selection to some extent.
- d) **evolution is hampered by the variability of the processes of birth and death.**

Q11. Which of the following statements will not weaken the 'previous view, directly descended from Thomas Malthus' mentioned in the last para of the passage?

a) **The variance of human fertility is independent of cultural and societal influence.**

b) **Culture and societal conventions influence individual choice.**

c) Human fertility is neither constrained by nor correlated to cultural and societal factors.

d) **Human physiology is naturally variable.**

Q12. A benefit of the ‘theoretical cohesion’ mentioned in the penultimate para of the passage is that

- a) **it connects evolutionary framework with the subject of human fertility.**
- b) **it aids in the process of practical decision-making.**
- c) it helps differentiate between policies that are effective and policies that aren’t.
- d) **it has offered biological anthropology the spotlight it deserves.**

DIRECTIONS *for questions 13 to 16*: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Troels Schönfeldt can trace his path to becoming a nuclear energy entrepreneur back to 2009, when he and other young physicists at the Niels Bohr Institute in Copenhagen started getting together for an occasional “beer and nuclear” meetup... The “nuclear” part was usually a session about their options for fighting two of humanity’s biggest problems: global poverty and climate change. “If you want poor countries to become richer,” says Schönfeldt, “you need a cheap and abundant power source.” But if you want to avoid spewing out enough extra carbon dioxide to fry the planet, you need to provide that power without using coal and gas.

It seemed clear to Schönfeldt and the others that the standard alternatives simply wouldn’t be sufficient. Wind and solar power by themselves couldn’t offer nearly enough energy, not with billions of poor people trying to join the global middle class. Yet conventional nuclear reactors — which could meet the need, in principle — were massively expensive, potentially dangerous and anathema to much of the public. And if anyone needed a reminder of why, the catastrophic meltdown at Japan’s Fukushima Daiichi plant came along to provide it in March 2011.

On the other hand, says Schönfeldt, the worldwide nuclear engineering community was beginning to get fired up about unconventional reactor designs — technologies sidelined 40 or 50 years before, but that might have a lot fewer problems than existing reactors. And the beer-and-nuclear group found that one such design, the molten salt reactor, had a simplicity, elegance, and, weirdness that especially appealed.

The weird bit was that word “molten,” says Schönfeldt: Every other reactor design in history had used fuel that’s solid, not liquid. This thing was basically a pot of hot nuclear soup. The recipe called for taking a mix of salts — compounds whose molecules are held together electrostatically, the way sodium and chloride ions are in table salt — and heating them up until they melted. This gave you a clear, hot liquid that was about the consistency of water. Then you stirred in a salt such as uranium tetrafluoride, which produced a lovely green tint, and let the uranium undergo nuclear fission right there in the melt — a reaction that would not only keep the salts nice and hot, but, could power a city or two besides.

…Weird or not, molten salt technology was viable; the Oak Ridge National Laboratory in Tennessee had successfully operated a demonstration reactor back in the 1960s. And more to the point, the beer-and-nuclear group realized, the liquid nature of the fuel meant that they could potentially build molten salt reactors that were cheap enough for poor countries to buy; compact enough to deliver on a flatbed truck; green enough to burn our existing stockpiles of nuclear waste instead of generating more — and safe enough to put in cities and factories. That’s because Fukushima-style meltdowns would be physically impossible in a mix that’s molten already. Better still, these reactors would be proliferation resistant, because their hot, liquid contents would be very hard for rogue states or terrorists to hijack for making nuclear weapons. Molten salt reactors might just turn nuclear power into the greenest energy source on the planet.

Q13. All the following are shortcomings with ‘standard alternatives’ mentioned in the first sentence of the second para of the passage EXCEPT:

- a) **There isn’t enough wind and solar power to fuel the global growth.**
- b) **Conventional nuclear reactors are expensive and dangerous.**
- c) Public opinion about nuclear energy is highly antagonistic.
- d) **There isn’t enough infrastructure to cope with the wind and solar power demand.**

Q14. It can be inferred from the last sentence of the penultimate para of the passage that

- a) **the nuclear fission reaction generated heat.**
- b) **uranium tetrafluoride is an environmental-friendly salt.**
- c) nuclear fission reaction can take place only at high temperatures.
- d) **it is practically feasible to transform heat generated from the nuclear fission reaction to other forms of energy.**

Q15. Which of the following can be inferred from the last para of the passage?

- a) **The owners of molten salt reactors may use the technology and process to make bombs.**
- b) **Unlike in conventional nuclear reactors, the molten salt reactors are hard to monitor for materials that can be used to make bombs on the side.**
- c) The proliferation resistance of molten salt reactors would be equivalent to the uranium/plutonium fuel cycles of conventional civilian nuclear reactors.
- d) **The contents of molten salt reactors cannot be easily extracted from the reactors.**

Q16. Which of the following statements, if true, will the author not use to support his central claim in the passage?

- a) **Molten salt reactors can use up nuclear waste produced by conventional nuclear plants.**
- b) **A chemical system can continuously extract reaction-slowing fission products from the molten fuel in molten salt reactors, thus increasing efficiency.**
- c) The molten core of the molten salt reactors would trap fission products far more securely than in solid-fuelled reactors.
- d) **A reactor which demonstrates the usage of molten salt as fuel will be ready in the next few years.**

Q17. **DIRECTIONS** *for questions 17 and 18*: The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

As long as we continue to talk and think as if the mental and the physical were separate metaphysical realms, the relation of the brain to consciousness will forever seem mysterious, and we will not have a satisfactory explanation of the relation of neuron firings to consciousness. The first step on the road to philosophical and scientific progress in these areas is to forget about the tradition of Cartesian dualism and just remind ourselves that mental phenomena are ordinary biological phenomena in the same sense as photosynthesis or digestion. We must stop worrying about how the brain could cause consciousness and begin with the plain fact that it does.

a) **That the brain creating consciousness is a normal biological phenomenon is an idea whose time has arrived.**

b) **The mysterious relation between the brain and the consciousness can be unravelled by accepting that all mental phenomena are ordinary biological phenomena.** ✓Your answer is correct

c) The brain and consciousness aren't two separate metaphysical realms but one and the same.

d) **Philosophical and scientific progress can only move forward if the brain and consciousness are seen as related to each other like any other biological phenomena.**

Q18. **DIRECTIONS** *for questions 17 and 18*: The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

On the Homa Peninsula in southwestern Kenya, east of Lake Victoria, lies an archaeological site where researchers have found some of humanity's oldest tools. These implements, known to experts as Oldowan – named after the Olduvai Gorge, an archaeologically rich site nearby in Tanzania – were shaped by our human ancestors into an impressive variety of wedges and hammers between 2.6 million and 1.6 million years ago, at the very start of the Stone Age. Studying this breakthrough technology, researchers have gained new insights into this Lower Paleolithic phase of human evolution. Long before widespread agriculture, the development of Oldowan led to better-fed, smarter hominins – and may well have enabled these largely itinerant early humans to begin establishing a sense of home.

a) **The Homa Peninsula in southwestern Kenya is home to the Oldowan tools which could be related to the turning point in human smartness.**

b) **The Oldowan tools enabled the Lower Paleolithic phase of human evolution possibly involving smarter hominins developing a sense of home.** ✓ Your answer is correct

c) The Oldowan implements establish at what point hominins were smart enough to start thinking of settling down.

d) **An impressive variety of implements helped hominins in the Homa Peninsula of Kenya with better nutrition that facilitated their evolution.**

Q19. DIRECTIONS *for questions 19 and 20*: The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. The crime series already had a cult following by that stage, but Netflix supercharged its popularity by putting the first three seasons up on its platform ahead of the fourth.

2. Not only could new viewers quickly get up to speed, but they could also consume Breaking Bad the same way Walter White's customers consumed his product.

3. Netflix has not only transformed our entertainment landscape but our social one, too.

4. In 2010, the company acquired the now-feted Breaking Bad, which had just finished its third season on AMC.

Q20. DIRECTIONS *for questions 19 and 20*: The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. While the private sector plays a pivotal role in advancing and maturing innovation, advancements are proprietary and are limited in scale.

2. There is a need and opportunity for more public-private partnerships that can bring the benefits of private sector technology and innovation to the public sector to achieve scale and impact.

3. Technological and operational innovations that directly address human, environmental and animal health issues have yet to gain momentum across industries.

4. Within this partnership, the private sector brings its market-driven approach and innovation in the development of treatments, while the public sector helps identify public health needs, inject funding, and shape incentives for private sector engagement.

Q21. DIRECTIONS *for question 21:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

The lyrebirds' performances are a classic example of an excessively complex trait or behavior that was likely shaped through sexual selection. Charles Darwin first introduced this theory in 1859 to explain why elaborate, seemingly non-adaptive characteristics, such as a peacock's tail, persist over generations. Virtually all birds make short calls to communicate. But Darwin argued that birdsong, which is more lyrical and complex, belongs almost exclusively to males, who use it to either attract mates or establish territory; the role of females, he believed, was to silently judge whose melody was most appealing.

a) **Darwin's sexual selection theory explains why and how excessively complex traits and behaviours like birdsongs persist over generations.**

b) **Birdsongs are an example of a complex trait shaped by sexual selection because males use birdsongs to attract mates and define territory.**

c) Lyrebirds' performances and the peacock's tail are complex traits that are shaped by selection.

d) **Only sexual selection can explain why elaborate, non-adaptive characteristics are passed down across generations.**

Q22. DIRECTIONS *for question 22:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. Scallop dredging, along with the broader bottom-trawling industry, targets marine life by dragging heavy gear along the seabed.

2. Each year, this fishery rakes the world's continental shelves, covering an estimated area equal to the size of Brazil, India and the second-largest country in Africa, the Democratic Republic of Congo, combined.

3. For decades, marine scientists have warned that bottom trawling destroys marine habitats.

4. Less publicized has been the impact on underwater cultural heritage, even though it's clear that the industry has long dredged history along with fish.

Q23. DIRECTIONS *for questions 23 and 24:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.
Sentence: ***The concept caught on like wildfire in the 2010s.***

Paragraph: It's useful to understand the origin of nudging. _____(1)_____. The concept comes from the field of behavioral economics known as nudge theory, laid out in the aptly named 2008 book Nudge by economist Richard Thaler and legal scholar Cass Sunstein. _____(2)_____. Nudge theory suggests that subtle cues can encourage good human behavior without the need for coercive policies that limit choice or economic penalties that raise the cost of bad behavior. _____(3)_____. To nudge customers to eat better, for example, a restaurant might organize its menu by listing healthy options first, and bury unhealthy ones at the bottom. _____(4)_____. Governments sought to incorporate nudges into their policymaking. And some major companies hired behavioral economists to figure out how to change consumer behavior and improve employee performance.

- a) **Option 1**
- b) **Option 2**
- c) **Option 3** ✗ Your answer is incorrect
- d) **Option 4**

Q24. DIRECTIONS *for questions 23 and 24:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.
Sentence: ***The future is more mysterious than ever.***

Paragraph: _____(1)_____. Climate change that disrupts agricultural cycles, coupled with information technology advancing at warp speed, is simply going to unravel political systems across the globe that cannot measure up. _____(2)_____. It is getting increasingly harder to be a dictator these days, even as the weakness or absence of bureaucratic institutions make democratic change problematic in many countries. _____(3)_____. The poet John Keats urged us to be "content with half knowledge," which echoes Tolstoy's observation that much of what will happen in international affairs remains "inaccessible to the human mind." _____(4)_____. But neither should we lose hope.

a) **Option 1**

b) **Option 2**

c) **Option 3**

d) **Option 4** ✘ Your answer is incorrect